

A Non-Dual Positive Subcase of the 2-Local Triple-Homomorphism Problem

Compact C*-algebra domains

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Abstract

Burgos, Fernandez-Polo, Garces, and Peralta ask whether every 2-local triple homomorphism between C*-algebras is linear, and prove this when the domain is a von Neumann algebra (more generally, a JBW*-triple). We prove a non-dual subcase: if $A = \bigoplus_{i \in I}^{c_0} \mathcal{K}(H_i)$ is a compact C*-algebra and F is any JB*-triple, then every 2-local triple homomorphism $T : A \rightarrow F$ is linear and hence a triple homomorphism. The mechanism is a row-column factorization on rank-one operators followed by singular-value decomposition. The general problems remain open.

1 Source problem and status

Every C*-algebra is a JB*-triple for

$$\{a, b, c\} = \frac{1}{2}(ab^*c + cb^*a).$$

A map $T : E \rightarrow F$, not assumed linear or continuous, is a *2-local triple homomorphism* if for every $x, y \in E$ there is a bounded linear triple homomorphism $\Phi_{x,y} : E \rightarrow F$ satisfying $\Phi_{x,y}(x) = T(x)$ and $\Phi_{x,y}(y) = T(y)$.

Problem 1.1 of [1] asks:

Is every 2-local triple homomorphism between C*-algebras automatically linear?

Problem 1.2 asks the same for arbitrary JB*-triples. The source paper proves linearity when the domain is a JBW*-triple. The result below covers a class of non-dual domains. Here “compact C*-algebra” means a C*-algebra of the form $\bigoplus_{i \in I}^{c_0} \mathcal{K}(H_i)$.

In a recent contribution, we establish a generalization of the Kowalski-Słodkowski theorem for 2-local $*$ -homomorphisms on von Neumann algebras, showing that every (not necessarily linear nor continuous) 2-local $*$ -homomorphism from a von Neumann algebra or from a compact C^* -algebra into a C^* -algebra is linear and a $*$ -homomorphism. In the Jordan setting, it is proved that every 2-local Jordan $*$ -homomorphism from a JBW^* -algebra into a JB^* -algebra is linear and a Jordan $*$ -homomorphism (cf. [14]).

Every C^* -algebra A admits a ternary product given by

$$\{a, b, c\} := \frac{1}{2}(ab^*c + cb^*a) \quad (a, b, c \in A).$$

A linear map Φ between C^* -algebras A and B satisfying $\Phi(\{a, b, c\}) = \{\Phi(a), \Phi(b), \Phi(c)\}$, is called a *triple homomorphism*. A *2-local triple homomorphism* between A and B is a not necessarily linear nor continuous map $T : A \rightarrow B$ such that for every $a, b \in A$, there exists a triple homomorphism $\Phi_{a,b} : A \rightarrow B$ with $\Phi_{a,b}(a) = T(a)$ and $\Phi_{a,b}(b) = T(b)$. Motivated by the above commented Kowalski-Słodkowski theorem for von Neumann algebras, it seems natural to consider the following independent problem:

Problem 1.1. *Is every 2-local triple homomorphism between C^* -algebras (automatically) linear?*

It should be noted here that, even in the case of von Neumann algebras, the proofs and arguments given in the study of 2-local $*$ -homomorphisms [14], are no longer valid when considering Problem 1.1, because triple homomorphisms between C^* -algebras do not preserve the natural partial order given by the positive cone in a C^* -algebra.

Problem 1.1 can be posed in the more general setting of JB^* -triples. Let E and F be two JB^* -triples (see subsection 1.1 for definitions). A linear map $\Phi : E \rightarrow F$ which preserves the triple products is called a *triple homomorphism*. A (not necessarily linear nor continuous) mapping $T : E \rightarrow F$ is said to be a *2-local triple homomorphism* if for every $a, b \in E$ there exists a bounded (linear) triple homomorphism $\Phi_{a,b} : E \rightarrow F$, depending on a and b ,

Source PDF, page 2: Problem 1.1 and adjacent scope context.

2 Proof intuition

The source theorem cannot simply be applied to $\mathcal{K}(H)$: for infinite-dimensional H , the compact operators are not a dual Banach space. However, $\mathcal{K}(H)$ is densely assembled from rank-one operators. If one vector in a rank-one tensor is held fixed, the resulting row or column is a Hilbert space and a JBW^* -subtriple. The source theorem therefore makes T linear on every row and every column.

After applying a functional $\varphi \in F^*$, these two families of linearity statements say that

$$(\xi, \eta) \longmapsto \varphi(T(\theta_{\xi, \eta}))$$

is sesquilinear. It consequently defines a linear functional on finite-rank operators. A singular-value decomposition writes a finite-rank operator as a linear combination of mutually orthogonal rank-one tripotents. The finite orthogonal-tripotent lemma in the source paper then identifies this functional with $\varphi \circ T$ and, crucially, bounds it by the operator norm rather than the trace norm. It extends to $\mathcal{K}(H)$, proving that every scalarization of T is linear. Finally, norm approximation by finitely many elementary summands handles a c_0 -sum.

3 The elementary case

We use the following consequences of 2-locality proved in [1, Section 2]. For arbitrary JB*-triples E, F , every 2-local triple homomorphism $T : E \rightarrow F$ is complex homogeneous and 1-Lipschitz, and if $e_1, \dots, e_n$ are mutually orthogonal tripotents, then

$$T \left(\sum_{j=1}^n \lambda_j e_j \right) = \sum_{j=1}^n \lambda_j T(e_j) \quad (\lambda_j \in \mathbb{C}). \quad (1)$$

The same paper proves that a 2-local triple homomorphism whose domain is a JBW*-triple is linear [1, Theorem 3.8].

For $\xi, \eta \in H$, let

$$\theta_{\xi, \eta}(z) = \langle z, \eta \rangle \xi \quad (z \in H),$$

using the convention that the inner product is linear in its first variable.

Lemma 3.1. *Let $T : \mathcal{K}(H) \rightarrow F$ be a 2-local triple homomorphism. For every unit vector η , T is linear on the column*

$$C_\eta = \{\theta_{\xi, \eta} : \xi \in H\},$$

and for every unit vector ξ , it is linear on the row

$$R_\xi = \{\theta_{\xi, \eta} : \eta \in H\}.$$

Proof. Both C_η and R_ξ are closed JB*-subtriples of $\mathcal{K}(H)$. For example,

$$\{\theta_{x, \eta}, \theta_{y, \eta}, \theta_{z, \eta}\} = \frac{1}{2}(\langle z, y \rangle \theta_{x, \eta} + \langle x, y \rangle \theta_{z, \eta}),$$

so C_η is the standard Hilbertian Cartan factor; the row calculation is the same. A Hilbertian Cartan factor is reflexive and is a JBW*-triple.

For two elements of C_η , restrict an implementing triple homomorphism for T on $\mathcal{K}(H)$ to C_η . Thus $T|_{C_\eta}$ is itself a 2-local triple homomorphism. The JBW*-domain theorem [1, Theorem 3.8] makes it linear. The same argument applies to R_ξ . $\square$

Theorem 3.2. *Let H be a complex Hilbert space and F a JB*-triple. Every 2-local triple homomorphism $T : \mathcal{K}(H) \rightarrow F$ is linear and a triple homomorphism.*

Proof. Fix $\varphi \in F^*$ and set

$$b_\varphi(\xi, \eta) = \varphi(T(\theta_{\xi, \eta})).$$

By Lemma 3.1, b_φ is linear in ξ and conjugate-linear in η . Homogeneity deals with non-unit fixed vectors. The Lipschitz estimate for T , together with $T(0) = 0$, gives

$$|b_\varphi(\xi, \eta)| \leq \|\varphi\| \|\xi\| \|\eta\|. \quad (2)$$

Hence b_φ induces a well-defined algebraic linear functional L_φ on the finite-rank operators by

$$L_\varphi \left(\sum_{k=1}^m \theta_{\xi_k, \eta_k} \right) = \sum_{k=1}^m b_\varphi(\xi_k, \eta_k).$$

Indeed, the canonical map from the algebraic tensor product $H \otimes \overline{H}$ onto the finite-rank operators is an algebraic isomorphism, and the sesquilinearity just proved is precisely the balancing needed for this definition.

Let x be finite-rank. Choose a singular-value decomposition

$$x = \sum_{j=1}^n s_j \theta_{\xi_j, \eta_j},$$

where $s_j \geq 0$ and both (ξ_j) and (η_j) are orthonormal. The operators $e_j = \theta_{\xi_j, \eta_j}$ are mutually orthogonal rank-one tripotents. Consequently, (1) yields

$$T(x) = \sum_{j=1}^n s_j T(e_j), \quad L_\varphi(x) = \varphi(T(x)).$$

It follows that

$$|L_\varphi(x)| = |\varphi(T(x))| \leq \|\varphi\| \|T(x)\| \leq \|\varphi\| \|x\|. \quad (3)$$

Thus L_φ extends uniquely to a bounded linear functional on $\mathcal{K}(H)$. If x_n are finite-rank and $x_n \rightarrow x$ in norm, continuity of T and L_φ gives

$$\varphi(T(x)) = \lim_n \varphi(T(x_n)) = \lim_n L_\varphi(x_n) = L_\varphi(x).$$

Therefore $\varphi \circ T$ is linear for every $\varphi \in F^*$. Since F^* separates points, T is additive; it was already complex homogeneous, so it is linear. Finally, a linear 2-local triple homomorphism preserves the triple product by the polarization argument in [1, Section 2]. $\square$

4 Compact C^* -algebras

Theorem 4.1 (compact-domain subcase). *Let*

$$A = \bigoplus_{i \in I}^{c_0} \mathcal{K}(H_i)$$

and let F be a JB^ -triple. Every 2-local triple homomorphism $T : A \rightarrow F$ is linear and a triple homomorphism.*

Proof. Write $A_i = \mathcal{K}(H_i)$. Restricting the implementing triple homomorphisms shows that $T_i = T|_{A_i}$ is a 2-local triple homomorphism. By Theorem 3.2, each T_i is linear.

If $i \neq j$, $x \in A_i$, and $y \in A_j$, choose finite singular-value truncations $x_n \rightarrow x$ and $y_n \rightarrow y$. Each x_n and y_n is a finite linear combination of rank-one tripotents, and all tripotents appearing in the two different summands are mutually orthogonal. Applying (1) and then passing to the norm limit gives

$$T(x + y) = T(x) + T(y). \quad (4)$$

Induction gives the analogous identity for every finite set of distinct summands.

For $a = (a_i)_{i \in I} \in A$, its finite coordinate truncations converge to a in norm. Since T is Lipschitz,

$$T(a) = \lim_J \sum_{i \in J} T_i(a_i),$$

where J ranges over finite subsets of I . For $a, b \in A$, linearity of every T_i now gives

$$T(a + b) = \lim_J \sum_{i \in J} T_i(a_i + b_i) = T(a) + T(b).$$

Together with complex homogeneity, this proves linearity. The polarization argument cited in Theorem 3.2 proves that T is a triple homomorphism. $\square$

5 Verification, limitations, and novelty check

Internal checks. The proof uses only four inputs from the source paper: homogeneity, 1-Lipschitz continuity, linearity on finite combinations of mutually orthogonal tripotents, and the JBW*-domain theorem. The two points most deserving independent review are the row/column restriction in Lemma 3.1 and the operator-norm estimate (3). Notice that the latter estimate is not inferred from the sesquilinear-form bound (2); it comes from 2-locality and orthogonal singular vectors. This is what forces extension to the operator-norm completion $\mathcal{K}(H)$.

Scope. Infinite-dimensional $\mathcal{K}(H)$ is not a JBW*-triple, so Theorem 3.2 is not a restatement of the source theorem. The result does not cover arbitrary C*-algebras, and it gives no bidual-extension principle for a general 2-local map. Problems 1.1 and 1.2 therefore remain open outside the stated compact-domain class.

Bounded novelty check. On 22 July 2026 we searched the run’s parsed arXiv corpus for the exact source questions and for “2-local triple homomorphism”, and performed bounded web/arXiv searches combining that phrase with “compact C*-algebra”, “ $K(H)$ ”, and “real rank zero”. The searches returned the source paper and papers citing it, but no statement of Theorem 4.1. The source introduction mentions a compact-domain theorem for 2-local **-homomorphisms*, not for triple homomorphisms. This is evidence only; the present deduction has not been peer reviewed and the novelty confidence is moderate.

Recommendation. Promote as a likely-valid partial result, subject to expert review of the three checkpoints listed in the accompanying README. Do not count it as a full answer to either source problem.

References

- [1] M. J. Burgos, F. J. Fernandez-Polo, J. J. Garcés, and A. M. Peralta, *2-local triple homomorphisms on von Neumann algebras and JBW*-triples*, J. Math. Anal. Appl. **426** (2015), 43–63; arXiv:1405.3836, doi:10.1016/j.jmaa.2014.12.058.